TEXORA · LECTURE SHEET

Humidification in Textile Mills

Cotton is hygroscopic: its moisture, strength, flexibility and static behaviour all follow the humidity of the room air. Humidification control keeps each department at the relative humidity its process needs. This sheet covers the department standards, the psychrometric calculation of water to be added, and the regain effects that show up in production and in the accounts.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 13 · SPECIALIZATION — YARN MANUFACTURING

1. Why humidity is a process variable

Fibres in a department move toward moisture equilibrium with the air. At low RH cotton loses regain: fibres become brittle, fly liberation rises, static electricity builds (fibres repel, lap onto rollers, attract dust), and ends down climb — the "low RH episodes" bar in the ring-frame Pareto of Topic 8. At excessive RH the penalties reverse: rollers lap with sticky fibre, metal corrodes, mildew risk grows, and yarn packages gain water that the invoice calculation (YM-3207 Topic 6) will later remove. Each department therefore has an RH standard matched to its material state, held by the humidification plant under automatic control.

DEPARTMENT	TYPICAL RH STANDARD (%)	REASON FOR THE LEVEL
Mixing / blowroom	55–60	Opening without brittle fibre damage; dust suppression
Carding	50–55	Web cohesion without lapping on rollers
Drawframe / simplex	55–60	Drafting control; static suppression
Ring spinning	55–60 (up to 65 for fine counts)	Ends down minimum; fibre flexibility at the balloon
Winding	60–65	Yarn relaxes to delivery regain; fewer breaks at clearers
(For contrast) Weaving	70–80	Warp sized yarn needs maximum flexibility

2. The psychrometric calculation

The water a plant must evaporate follows from the humidity ratio W (kg water per kg dry air) at the outside/supply condition and at the department target, read from the psychrometric chart or computed from vapour pressures:

$$\text{Water to add (kg/h)} = \text{dry-air flow (kg/h)} \times (W_{target} - W_{supply})$$

WORKED EXAMPLE — HUMIDIFICATION LOAD OF A SPINNING SHED

Department volume 5400 m³ (60 m × 30 m × 3 m); ventilation 6 air changes per hour; air density (dry-air basis) 1.13 kg/m³. Air is supplied at 30 °C and 40% RH (W = 10.6 g/kg); the department target is 30 °C and 65% RH (W = 17.4 g/kg).

Air flow = 5400 × 6 = 32,400 m³/h → dry-air flow = 32,400 × 1.13 = **36,612 kg/h**.

Moisture deficit = 17.4 − 10.6 = 6.8 g/kg = 0.0068 kg/kg.

Water to evaporate = 36,612 × 0.0068 = **249 kg/h**.

The spray system supplies 340 kg/h of water, so its evaporation efficiency = 249/340 × 100 = **73%** — the rest drains away unevaporated, which is normal for spray humidification and is why drain water is recirculated after filtration and treatment.

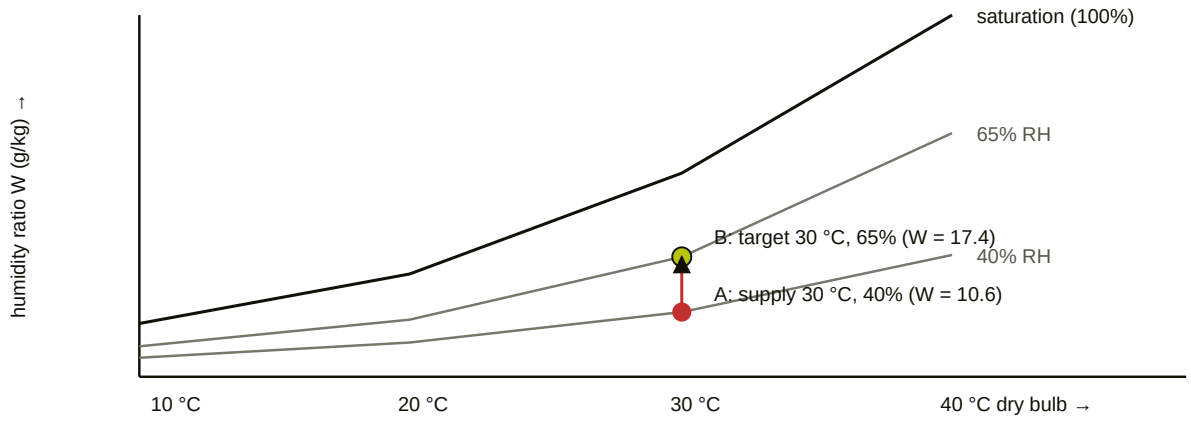


Figure 1 — Psychrometric chart in sketch form. Humidification at constant temperature moves the air state vertically, A → B: W rises from 10.6 to 17.4 g/kg at 30 °C. (Curves are schematic, values from the worked example.)

3. Regain effects the plant is really controlling

WORKED EXAMPLE — WATER CARRIED BY THE PRODUCT

Cotton yarn at equilibrium with 55% RH holds about 7.0% regain; at 75% RH about 9.3%.

One tonne of oven-dry fibre weighs 1000 × 1.070 = **1070 kg** in the first room and 1000 × 1.093 = **1093 kg** in the second — a difference of **23 kg of water per tonne**.

> This water is not cosmetic: it changes fibre flexibility and strength (cotton is stronger wet than dry, up to a point), it changes measured weights at every process test, and it is exactly the mass that count correction and invoice weight (YM-3207 Topic 6) exist to normalise. Departments that let RH wander are, in effect, letting the product's test weights wander.

4. Plant types and control practice

- **Air washer / spray humidifiers** — water sprayed into the air stream; evaporation efficiency as computed above; also washes dust from the air. Needs water treatment to prevent microbial growth and nozzle scaling.
- **Atomising nozzles (compressed-air or high-pressure)** — fine mist directly in the department; high evaporation efficiency; local control per zone.
- **Steam humidification** — precise but energy-costly; used for small or critical zones.
- **Control instrumentation:** RH sensors per department logged continuously (aspirated psychrometers as reference); alarms on excursion, because a failed plant shows in ends down within the hour. Sensor calibration is part of the machinery audit (Topic 12) — an RH sensor reading 5% low will hold the whole shed 5% dry while reporting perfection.

Glossary

Relative humidity (RH) — actual vapour pressure as % of saturation vapour pressure at the same temperature. **Humidity ratio (W)** — mass of water vapour per unit mass of dry air. **Dry bulb temperature** — ordinary air temperature. **Psychrometric chart** — the chart relating temperature, RH and humidity ratio of moist air. **Air changes per hour** — department volume replaced per hour by the ventilation system. **Evaporation efficiency** — the fraction of sprayed water that actually evaporates into the air. **Hygroscopic** — absorbing and releasing moisture with the surrounding air, as all natural fibres do.

Key points

• RH standards rise along the process: carding 50–55%, ring spinning 55–60%, winding 60–65%, weaving 70–80%.
• Humidification load = dry-air flow × ΔW: 36,612 kg/h × 6.8 g/kg = 249 kg/h in the example; spray evaporation efficiency 73%.
• On the psychrometric chart, humidification at constant temperature is a vertical move, A → B.
• Regain follows RH: one tonne of dry cotton carries 1070 kg at 55% RH equilibrium but 1093 kg at

75% — 23 kg of water that tests and invoices must normalise.

• RH excursions appear within the hour as ends down and fly; sensors are logged, alarmed and audited.

Practice

1. A shed of 3600 m³ has 5 air changes/h and air density 1.13 kg/m³. Supply air W = 9.0 g/kg, target W = 16.0 g/kg. Calculate the water to be evaporated per hour.
2. A spray plant supplies 260 kg/h of water and evaporates 182 kg/h. Calculate its evaporation efficiency and the drain flow.
3. Ring-frame ends down double during a dry afternoon while all machine settings test normal. Using this sheet, explain the mechanism and name the record you would check to confirm it.

Answer key

1. Air flow = 3600 × 5 = 18,000 m³/h → dry air = 18,000 × 1.13 = 20,340 kg/h. ΔW = 7.0 g/kg. Water = 20,340 × 0.007 = **142 kg/h**.
2. Efficiency = 182/260 × 100 = **70%**. Drain = 260 − 182 = **78 kg/h** (recirculated after treatment).
3. Falling RH dries the fibres: regain drops, fibres turn brittle, static builds and fly liberation rises, so ends break at drafting and at the balloon. Confirm from the department's continuous RH log — a dip matching the afternoon's break pattern confirms a climate cause rather than a machine one.